

README.md — Task 6

List and Dictionary Comprehensions

Data Science & Data Analytics Using Python Internship

Project Overview

This project is part of the Data Science & Data Analytics Using Python Internship — Task 6. The task focuses on using List Comprehensions and Dictionary Comprehensions in Python for efficient data processing, filtering, transformation, cleaning, and classification.

Objective

To understand how comprehensions can be used to write concise and efficient Python code for basic data-processing and data-analysis tasks.

Tools & Technologies

- Python
- Jupyter Notebook
- VS Code
- CSV datasets

Topics Covered

- List Comprehension
- Filtering with conditions
- Data transformation
- String processing
- Nested List Comprehension
- Dictionary Comprehension
- Dictionary filtering
- Student marks and grade classification
- Product price transformation
- Data cleaning
- Practical data-processing examples

Project Files

```
Task_6/  
■  
■■■ Task_6_Complete_Assessment.ipynb  
■■■ Task_6_Complete_Assessment.py  
■■■ student_marks.csv  
■■■ product_dataset.csv  
■■■ README.md
```

Example — List Comprehension

```
numbers = [1, 2, 3, 4, 5]  
squares = [n ** 2 for n in numbers]  
print(squares)
```

Output: [1, 4, 9, 16, 25]

Example — Filtering

```
even_numbers = [n for n in numbers if n % 2 == 0]  
print(even_numbers)
```

Output: [2, 4]

Example — Dictionary Comprehension

```
students = {  
    "Rahul": 85,  
    "Priya": 92,  
    "Aman": 67
```

```
}

passed_students = {
    name: marks
    for name, marks in students.items()
    if marks >= 60
}

print(passed_students)
```

Practical Applications

The project demonstrates comprehensions using student marks, grade classification, passed-student filtering, product prices, discount calculations, premium-product filtering, text cleaning, and data transformation.

How to Run

Using Jupyter Notebook:

1. Open Jupyter Notebook.
2. Open Task_6_Complete_Assessment.ipynb.
3. Run each cell sequentially.

Using Python:

```
python Task_6_Complete_Assessment.py
```

Learning Outcomes

After completing this task, you will be able to create lists using list comprehensions, filter data using conditions, transform values efficiently, create dictionaries using dictionary comprehensions, process student and product data, apply comprehensions to basic data-analysis tasks, and write more concise Python programs.

Top 5 Skills

1. Python Programming
2. List & Dictionary Comprehensions
3. Data Filtering & Transformation
4. Data Processing
5. Problem Solving

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Task Status

Task 6 — Completed ■